

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12403563
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Nakamura; Tatsuya et al.

Power tool

Abstract

A power tool includes a spindle, a housing, a cover body configured to be removably mounted to the housing, a first engagement part provided on or in the cover body, and a second engagement part provided on or in the housing. The cover body includes an upper plate part and an outer peripheral part. The second engagement part is movable between an engagement position and a disengagement position. The first and second engagement parts are configured to restrict rotation of the cover body around the drive axis relative to the housing by engaging with each other. The first engagement part is (i) at a same location as or radially outward of the outer peripheral part of the cover body in a radial direction orthogonal to the drive axis, and (ii) between an upper end and a lower end of the cover body in the up-down direction.

Inventors: Nakamura; Tatsuya (Anjo, JP), Kaneko; Hiroki (Anjo, JP), Tanaka; Yunosuke (Anjo, JP), Matsuoka; Yukiya (Anjo, JP), Kawakami; Takahiro (Anjo, JP), Kotsuji; Takafumi (Anjo, JP)

Applicant: MAKITA CORPORATION (Anjo, JP)

Family ID: 91026405

Assignee: MAKITA CORPORATION (Anjo, JP)

Appl. No.: 18/516436

Filed: November 21, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240173821 A1	May. 30, 2024

Foreign Application Priority Data

JP	2022-187779	Nov. 24, 2022
JP	2022-187780	Nov. 24, 2022

Publication Classification

Int. Cl.: B24B55/05 (20060101); B24B23/02 (20060101); B25F5/02 (20060101)

U.S. Cl.:

CPC B24B55/052 (20130101); B25F5/02 (20130101); B24B23/028 (20130101)

Field of Classification Search

CPC: B24B (55/052); B24B (23/028); B25F (5/02)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4924635	12/1989	Rudolf	451/344	B24B 55/052
5384985	12/1994	Jacobsson	451/344	B24B 55/052
5386667	12/1994	Hausslein	451/344	B24B 55/052
5766062	12/1997	Edling	451/454	B23Q 11/06
6949017	12/2004	Koschel	451/451	B24B 23/02
7175513	12/2006	Folin	451/451	B24B 55/052
7909680	12/2010	Boeck	451/454	B24B 55/052
8282446	12/2011	Sulea	451/451	B24B 55/052
8939816	12/2014	Chen	451/359	B24B 23/028
9079291	12/2014	Ikuta	N/A	B24B 23/028
9358664	12/2015	Ikuta	N/A	B24B 55/052
2004/0203330	12/2003	Kleider	451/359	B24B 23/028
2005/0215186	12/2004	Stierle	451/359	B24B 55/052
2006/0052041	12/2005	Wiker	451/451	B24B 55/052
2008/0153404	12/2007	Schmidberger-Brinek	451/359	B24B 55/052
2010/0105300	12/2009	Esenwein	451/452	B24B 55/052
2010/0178857	12/2009	Esenwein	451/451	B24B 55/052
2012/0052779	12/2011	Timcke	451/451	B24B 23/022
2012/0270480	12/2011	Rohde	451/451	B24B 55/052
2012/0295524	12/2011	Numata	451/363	B24B 23/028
2013/0000935	12/2012	Kelleher	173/46	B23Q 11/08
2016/0016285	12/2015	Cooksey	451/359	B24B 23/028
2016/0039068	12/2015	Morris	29/434	B24B 23/02
2016/0297052	12/2015	Aiken	N/A	B24B 23/028
2017/0320192	12/2016	Herchek	N/A	B24B 23/028
2018/0236634	12/2017	Chen	N/A	B24B 55/052
2024/0173839	12/2023	Kawakami	N/A	B25F 5/02
2024/0198482	12/2023	Bowers	N/A	B27G 19/04

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
107520758	12/2016	CN	N/A
2189244	12/2009	EP	B24B 23/02

Primary Examiner: Palmer; Lucas E. A.

Attorney, Agent or Firm: Oliff PLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application claims priority to Japanese patent application Nos. 2022-187779 and 2022-187780 filed on Nov. 24, 2022, the contents of which are hereby fully incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a power tool. More specifically, the present disclosure relates to a power tool that includes a cover that partially covers a tool accessory mounted to a spindle.

BACKGROUND

(3) Some power tools (e.g., power tools that rotate or oscillate a tool accessory) are provided with a cover that partially covers a tool accessory mounted to a lower end portion of a spindle in order to suppress scattering of dust and sparks that are generated during a processing operation. In order to facilitate handling of a power tool in a narrow space, it is preferable that the power tool is small in size in an axial direction of the spindle. Accordingly, in order to reduce the size of the spindle in the axial direction, various improvements in a mounting structure of the cover have been proposed. For example, European Patent No. 2189244 discloses a grinder that has a lever that is disposed radially outward of a protection hood (cover) and configured to engage with the protection hood.

SUMMARY

(4) In the above-described grinder, the protection hood has a disc-like part that is placed above a tool accessory and an edge part that is connected to an outer edge of the disc-like part and surrounds an edge of the tool accessory. The lever is disposed radially outward of the protection hood and configured to engage with a recess that is formed in a portion of the edge part of the protection hood that is below the tool accessory. Such a cover mounting structure leaves room for further improvement.

(5) It is accordingly a non-limiting object of the present disclosure to provide an improvement in a structure of mounting a cover for a tool accessory to a power tool.

(6) A non-limiting aspect of the present disclosure herein provides a power tool that includes a spindle, a housing, a cover body, a first engagement part and a second engagement part. The power tool of this aspect may include a rotary tool (e.g., a grinder, a cutter and, a circular saw) that is configured to rotate a tool accessory around a drive axis, and an oscillating tool (a so-called multi-tool) that is configured to oscillate (reciprocally pivot) a tool accessory about a drive axis.

(7) The spindle extends along a drive axis that defines an up-down direction of the power tool. The spindle has a lower end portion that is configured such that a tool accessory is removably mounted thereto. The housing houses the spindle with the lower end portion of the spindle exposed to an outside of the housing. The cover body is configured to be removably mounted to the housing to partially cover the tool accessory mounted to the lower end portion of the spindle. The cover body includes an upper plate part and an outer peripheral part (outer circumferential part). The upper plate part is configured to be arranged above the tool accessory. The outer peripheral part extends along an outer edge of the upper plate part and protrudes downward from the outer edge of the upper plate part. The first engagement part is provided on or in the cover body. The second engagement part is provided on or in the housing. The second engagement part is movable between

(a) an engagement position, at which the second engagement part engages with the first engagement part, and (b) a disengagement position, at which the second engagement part is not engageable with the first engagement part. The first engagement part and the second engagement part are configured to restrict rotation of the cover body around the drive axis relative to the housing by engaging with each other. The first engagement part is at a same location as the outer peripheral part of the cover body or radially outward of the outer peripheral part of the cover body in a radial direction that is orthogonal to the drive axis. The first engagement part is between upper end and a lower end of the cover body in the up-down direction. For example, if the first engagement part is a recess or hole formed in the outer peripheral part, the first engagement part may be regarded as being “at the same location as the outer peripheral part in the radial direction”. A position that is at a same level as the upper end and a position that is at a same level as the lower end in the up-down direction can be regarded as being “between the upper end and the lower end of the cover body in the up-down direction”.

(8) The power tool of this aspect is configured such that rotation of the cover body is restricted by the first engagement part provided on/in the cover body and the second engagement part provided on/in the housing that are engaged with each other. Further, the first engagement part is between the upper end and the lower end of the cover body in the up-down direction. Thus, the first engagement part does not protrude upward or downward from the cover body. Therefore, compared with a structure in which the first engagement part protrudes upward or downward from the cover body, the size of the entire power tool with the second engagement part in the up-down direction can be made smaller.

(9) Another non-limiting aspect of the present disclosure herein provides a power tool that includes a spindle, a housing, a cover body, a rail and a lever. An example of the power tool of this aspect is the same as an example of the power tool of the above-described aspect.

(10) The spindle extends along a drive axis that defines an up-down direction of the power tool. The spindle has a lower end portion that is configured such that a tool accessory is removably mounted thereto. The housing houses the spindle with the lower end portion of the spindle exposed to an outside of the housing. The cover body is configured to be removably mounted to the housing and to partially cover the tool accessory mounted to the lower end portion of the spindle. The cover body includes an upper plate part and an outer peripheral part (outer circumferential part). The upper plate part is configured to be arranged above the tool accessory. The outer peripheral part extends along an outer edge of the upper plate part and protrudes downward from the outer edge. The rail has an arcuate shape, and protrudes radially outward from the upper plate part or the outer peripheral part of the cover body. The lever is configured to be externally manipulated by a user. The lever is supported by the housing to be movable between (a) an engagement position, at which the lever engages with the rail, and (b) a disengagement position, at which the lever is not engageable with the rail. The rail and the lever are configured to restrict rotation of the cover body around the drive axis relative to the housing by engaging with each other.

(11) The power tool of this aspect is configured such that rotation of the cover body is restricted by the rail provided on the cover body and the lever provided on the housing. Further, the arcuate rail protrudes radially outward from the upper plate part or the outer peripheral part of the cover body. Therefore, compared with a structure in which the rail protrudes upward or downward from the cover body, the size of the entire power tool with the lever in the up-down direction can be made smaller.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a grinder of a first embodiment of the present disclosure.

- (2) FIG. 2 is a sectional view of the grinder.
- (3) FIG. 3 is a partial, enlarged view of FIG. 2.
- (4) FIG. 4 is a perspective view of a cover.
- (5) FIG. 5 is a perspective view of a front end portion of the grinder.
- (6) FIG. 6 is a partial, exploded perspective view of the grinder with a tool accessory and the cover removed therefrom.
- (7) FIG. 7 is a partial, left side view of the grinder with a lever placed in an engagement position and with the cover removed therefrom.
- (8) FIG. 8 is a sectional view showing a rail and the lever placed in the engagement position.
- (9) FIG. 9 is a partial, left side view of the grinder with the lever placed in a disengagement position and with the cover removed therefrom.
- (10) FIG. 10 is a sectional view showing the rail and the lever placed in the disengagement position.
- (11) FIG. 11 is a partial, left side view showing a grinder of a second embodiment, with a lever placed in the engagement position and with a cover removed therefrom.
- (12) FIG. 12 is a partial, sectional view showing a grinder of a third embodiment, with a lever placed in the engagement position.
- (13) FIG. 13 is a perspective view of a cover.
- (14) FIG. 14 is an enlarged perspective view showing a rail engagement part and its vicinity.
- (15) FIG. 15 is a partial, exploded perspective view of the grinder with the tool accessory and the cover removed therefrom.
- (16) FIG. 16 is a partial, sectional view of the grinder, with the lever placed in the disengagement position.
- (17) FIG. 17 is a partial, perspective view showing a grinder of a fourth embodiment.
- (18) FIG. 18 is a perspective view of a cover.
- (19) FIG. 19 is a bottom view of the grinder, with a lever placed in the engagement position.
- (20) FIG. 20 is a partial, exploded perspective view of the grinder with the tool accessory and the cover removed therefrom.
- (21) FIG. 21 is a bottom view of the grinder, with the lever placed in the disengagement position.
- (22) FIG. 22 is a partial, sectional view showing a grinder of a fifth embodiment, with a lever placed in the engagement position.
- (23) FIG. 23 is a perspective view of a cover.
- (24) FIG. 24 is a sectional view showing a rail and the lever placed in the engagement position.
- (25) FIG. 25 is a partial, perspective view of the grinder.
- (26) FIG. 26 is a partial, exploded perspective view of the grinder with the tool accessory and the cover removed therefrom.
- (27) FIG. 27 is a partial, sectional view of the grinder with the lever placed in the disengagement position.
- (28) FIG. 28 is a sectional view showing the rail and the lever placed in the disengagement position.
- (29) FIG. 29 is a partial, sectional view showing a grinder of a sixth embodiment, with a lever placed in the engagement position.
- (30) FIG. 30 is a partial, perspective view of the grinder.
- (31) FIG. 31 is a partial, exploded perspective view of the grinder with the tool accessory and a cover removed therefrom.
- (32) FIG. 32 is a sectional view showing a rail and the lever placed in the engagement position.
- (33) FIG. 33 is a partial, sectional view of the grinder with the lever placed in the disengagement position.
- (34) FIG. 34 is a sectional view showing the rail and the lever placed in the disengagement position.

(35) FIG. 35 is a partial, sectional view showing a grinder of a seventh embodiment, with a latch placed in the engagement position.

(36) FIG. 36 is a bottom view of the grinder, with the latch placed in the engagement position.

(37) FIG. 37 is a sectional view showing a cover and the latch placed in the disengagement position.

(38) FIG. 38 is a bottom view of the grinder with another cover mounted thereto.

(39) FIG. 39 is a bottom view of the grinder with yet another cover mounted thereto.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(40) In a non-limiting embodiment according to the present disclosure, the first engagement part may be configured as a protruding part that protrudes radially outward from the upper plate part or the outer peripheral part of the cover body. According to this embodiment, the freedom of design of the engagement structure of the first and second engagement parts can be enhanced.

(41) In addition or in the alternative to the preceding embodiment, the second engagement part may be a lever that is linearly movable substantially in parallel to the drive axis. According to this embodiment, rotation of the cover body can be effectively restricted by cooperation between the protruding part provided on the cover body and a simple lever provided on the housing.

(42) In addition or in the alternative to the preceding embodiments, the protruding part may have at least one recess or hole that extends in the up-down direction. The second engagement part may have a projection that is configured to engage with the recess or the hole when the second engagement part is at the engagement position. According to this embodiment, rotation of the cover body relative to the housing can be reliably restricted.

(43) In addition or in the alternative to the preceding embodiments, the lever may include (i) a manipulation part that is configured to be externally manipulated by a user above the housing, (ii) the projection, and (iii) a connection part that connects the manipulation part and the projection. According to this embodiment, the user can easily manipulate the lever.

(44) In addition or in the alternative to the preceding embodiments, the connection part may be at least partially within the housing. According to this embodiment, the possibility that the lever is affected by an unintended external force can be reduced.

(45) In addition or in the alternative to the preceding embodiments, the projection may be configured to be engaged with the protruding part of the cover body from below when the lever is at the engagement position. According to this embodiment, a user can move the lever from the engagement position to the disengagement position by pushing the manipulation part downward. The user can thus manipulate the lever more easily, compared with a structure that requires pulling the manipulation part upward.

(46) In addition or in the alternative to the preceding embodiments, the second engagement part may be a rotary lever that is configured to be externally manipulated by a user. According to this embodiment, the lever can be easily manipulated.

(47) In addition or in the alternative to the preceding embodiments, the second engagement part may be movable within or along a plane that is substantially orthogonal to the drive axis. According to this embodiment, the size of the second engagement part in the up-down direction can be reliably reduced.

(48) In addition or in the alternative to the preceding embodiments, the protruding part may have at least one recess that is recessed radially inward from an outer edge of the protruding part. The second engagement part may have a projection that is configured to be engaged with the recess when the second engagement part is at the engagement position. According to this embodiment, rotation of the cover body relative to the housing can be reliably restricted.

(49) In addition or in the alternative to the preceding embodiments, the protruding part may be configured to restrict movement of the cover body relative to the housing in a first direction, which is different from a circumferential direction around the drive axis, by engaging with one or both of the second engagement part and the housing. According to this embodiment, the protruding part

provided on the cover body can restrict (i) movement (rotation) of the cover body relative to the housing in the circumferential direction and (ii) movement of the cover body relative to the housing in the first direction that is different from the circumferential direction around the drive axis. Thus, the protruding part can perform two functions of (i) properly positioning the cover body in the circumferential direction and (ii) suppressing looseness or a play in the first direction. Therefore, compared with a case in which these two functions are achieved respectively by two separate (individual, discrete) structures, a space required in the power tool can be reduced and the power tool can be simplified in structure.

(50) In addition or in the alternative to the preceding embodiments, the housing may have a groove. The protruding part may be configured to restrict movement of the cover body relative to the housing in the first direction by being fitted in the groove. According to this embodiment, the cover body can be effectively positioned in the circumferential direction. In addition, the looseness or the play can be reduced by cooperation between the protruding part provided on the cover body and the second engagement part and the groove that are provided in the housing.

(51) In addition or in the alternative to the preceding embodiments, the second engagement part may be configured to restrict (i) rotation of the cover body relative to the housing and (ii) movement of the cover body relative to the housing in the first direction by pressing the protruding part against the housing when the second engagement part is at the engagement position. According to this embodiment, the cover body can be effectively positioned in the circumferential direction and the looseness or the play can be reduced by cooperation between the protruding part, the second engagement part and the housing.

(52) In addition or in the alternative to the preceding embodiments, the power tool may further include a biasing member that is configured to bias the second engagement part toward the engagement position. According to this embodiment, engagement between the first engagement part and the second engagement part can be maintained by the biasing force of the biasing member.

(53) In addition or in the alternative to the preceding embodiments, the first engagement part may be formed separately (discretely, individually) from the cover body and fixed to the cover body. According to this embodiment, the cover body and the first engagement part, which have been formed separately, can be fixed together afterwards. Such a configuration facilitates manufacturing, compared with a case in which the cover body and the first engagement part are integrally formed together from the beginning.

(54) In addition or in the alternative to the preceding embodiments, a lower end of the second engagement part may be at a same location as or above a lower end of the cover body in the up-down direction when the second engagement part is at the engagement position and when the second engagement part is at the disengagement position. According to this embodiment, the second engagement part does not protrude downward from the cover body. Therefore, the size of the entire power tool including the second engagement part can be reduced in the up-down direction. Further, the possibility that the second engagement part is affected by an unintended external force can be reduced.

(55) In addition or in the alternative to the preceding embodiments, the second engagement part may be between an upper end and the lower end of the cover body in the up-down direction both in the engagement position and in the disengagement position. According to this embodiment, the second engagement part does not protrude upward and downward from the cover body. Therefore, the possibility that the second engagement part is affected by an unintended external force can be further reduced.

(56) In addition or in the alternative to the preceding embodiments, the protruding part may be an arcuate rail. The rail may be configured to restrict movement of the cover body relative to the housing in a first direction that is different from a circumferential direction around the drive axis by being engaged with one or both of the lever and the housing. According to this embodiment, the cover body can be effectively positioned in the circumferential direction and the looseness or the

play can be reduced by cooperation between the rail and one or both of the lever and the housing. (57) Representative, non-limiting embodiments of the present disclosure are now specifically described with reference to the drawings. In the following embodiments, a hand-held electric disc grinder (hereinafter simply referred to as a grinder) is described as a representative example of a power tool of the present disclosure. The grinder is also a representative example of a rotary tool that is configured to rotationally drive the tool accessory.

First Embodiment

(58) A grinder **1A** according to a first embodiment is now described with reference to FIGS. **1** to **10**.

(59) The general structure of the grinder **1A** is now described. As shown in FIGS. **1** and **2**, the grinder **1A** includes a motor **21**, a spindle **25** that is operably coupled to the motor **21**, and a housing **10A** that houses the motor **21** and the spindle **25**. The housing **10A** has an elongate hollow body and forms an outer shell of the grinder **1A**. The motor **21** is arranged such that a rotational axis RX of an output shaft **215** of the motor **21** extends substantially in parallel to a longitudinal axis of the housing **10A**. The spindle **25** is disposed within one end portion of the housing **10A** in its longitudinal direction. The spindle **25** is supported within the housing **10A** so as to be rotatable around a drive axis DX. The drive axis DX crosses the rotational axis RX of the output shaft **215**. More specifically, the drive axis DX intersects the rotational axis RX at a substantially right angle.

(60) One axial end portion of the spindle **25** is exposed to the outside from the housing **10A**. A tool accessory **91** is removably mounted to this end portion of the spindle **25**. A portion of the tool accessory **91** is covered by a cover **5A** that is mounted (coupled) to the housing **10A**.

(61) When the spindle **25** is rotationally driven around the drive axis DX by the motor **21**, the tool accessory **91** is rotated, and a processing operation is performed on a workpiece. The tool accessory **91** that can be mounted to the grinder **1A** includes a grinding wheel, a cutting wheel, a blade, a rubber pad and a brush. The grinding wheel, the cutting wheel and the blade are non-limiting examples of the tool accessory **91** having a disc-like shape. A user selects the tool accessory **91** that is suitable for a desired operation and mounts it to the grinder **1A**. The grinder **1A** is capable of performing grinding, polishing, cutting or other similar operation on a workpiece, depending on the kind of the selected tool accessory **91**.

(62) The structure of the grinder **1A** is now described in detail. In the following description, for the sake of convenience, the extending direction of the drive axis DX is defined as an up-down direction of the grinder **1A**. In the up-down direction, the side on which the tool accessory **91** is located is defined as a lower side of the grinder **1A**, and the opposite side is defined as an upper side of the grinder **1A**. The extending direction of the rotational axis RX of the output shaft **215** is defined as a front-rear direction of the grinder **1A**. In the front-rear direction, the side on which the spindle **25** is located is defined as a front side of the grinder **1A**, and the opposite side is defined as a rear side of the grinder **1A**. A direction that is orthogonal to the up-down direction and the front-rear direction is defined as a left-right direction of the grinder **1A**. Further, any direction that is orthogonal to the drive axis DX is defined as a radial direction. In the radial direction, a direction away from the drive axis DX is defined as a radially outward direction, and a direction toward the drive axis DX is defined as a radially inward direction.

(63) First, the structure of the housing **10A** is described.

(64) As shown in FIGS. **1** and **2**, the housing **10A** includes a driving-mechanism housing part **11**, a motor housing part **13**, a handle part **15** and a controller housing part **17**, which are arranged in this order from the front to the rear. The driving-mechanism housing part **11** houses the spindle **25** and an intermediate shaft **23**. The driving-mechanism housing part **11** may also be referred to as a gear housing. The driving-mechanism housing part **11** forms a front end portion of the housing **10A**. The motor housing part **17** houses the motor **21** and a fan **22**. The handle part **15** is configured to be held by a user. The handle part **15** may also be referred to as a grip part. The controller housing part **17** houses a controller **29**. The controller housing part **17** forms a rear end portion of the housing

10A.

(65) Elements (mechanisms) disposed within the housing **10A** are now described.

(66) The output shaft **215** of the motor **21** is supported within the motor housing part **13** so as to be rotatable around the rotational axis RX. The fan **22** is fixed to a portion of the output shaft **215** that protrudes forward from a stator **211**. The fan **22** integrally rotates with the output shaft **215**.

(67) As shown in FIG. 3, the spindle **25** is supported within the driving-mechanism housing part **11** so as to be rotatable around the drive axis DX. A driven gear **250** is fixed onto an upper portion of the spindle **25**. A lower end portion of the spindle **25** protrudes downward from the housing **10A**. The lower end portion of the spindle **25** is configured as a tool mounting part **253** to which the tool accessory **91** is removably mounted (coupled). In this embodiment, an outer peripheral surface of the tool mounting part **253** is threaded and the tool accessory **91** can be fixed to the tool mounting part **253** by a lock nut **254**. The method of mounting the tool accessory **91** to the spindle **25** is not limited to this, but any known method may be employed.

(68) The grinder **1A** is provided with the cover **5A** for protecting a user from dust and sparks that are generated during a processing operation performed by the tool accessory **91** on a workpiece. The cover **5A** is configured to be removably mounted (coupled) to the housing **10A** such that the cover **5A** partially covers the tool accessory **91** mounted to the tool mounting part **253** of the spindle **25**. The cover **5A** may also be referred to as a wheel cover, a wheel guard, a disc cover, a protection cover or a protection hood. The cover **5A** is described below in further detail.

(69) As shown in FIG. 3, the intermediate shaft **23** is operably connected to the motor **21** and the spindle **25** and transmits rotational driving force of the motor **21** to the spindle **25**. Specifically, the intermediate shaft **23** is coaxial with the output shaft **215** of the motor **21** and supported within the driving-mechanism housing part **11** so as to be rotatable around the rotational axis RX. A rear end portion of the intermediate shaft **23** is engaged with a front end portion of the output shaft **215** via a connecting member, so that the intermediate shaft **23** integrally rotates with the output shaft **215**. A drive gear **230** is provided on a front end portion of the intermediate shaft **23**. The drive gear **230** is engaged with the driven gear **250**. Bevel gears are employed as the drive gear **230** and the driven gear **250**.

(70) With the above-described structure, when the motor **21** is driven, the spindle **25** is rotated around the drive axis DX via the intermediate shaft **23**, and the tool accessory **91** fixed to the tool mounting part **253** is integrally rotated with the spindle **25**. The rotational driving force of the motor **21** may however be transmitted to the spindle **25** via any mechanism other than the intermediate shaft **23**.

(71) As shown in FIG. 2, a switch **27** is housed within the handle part **15**. The switch **27** is for starting the motor **21**. In this embodiment, the motor **21** is energized and driven while the switch **27** is ON. A switch knob **271**, which is configured to turn On and OFF the switch **27** is provided on the housing **10A**. More specifically, the switch knob **271** is supported on an upper portion of the motor housing part **13** so as to be externally manipulated by the user and to be movable between an ON position and an OFF position. The switch knob **271** is normally held in the OFF position. The switch knob **271** is connected to the switch **27** via a connecting member **272**. The switch **27** is switched from OFF to ON when the user moves the switch knob **271** from the OFF position to the ON position.

(72) The controller **29**, which is configured to control operation of the grinder **1A**, is housed within the controller housing part **17**. The controller **29** includes a control circuit and is electrically connected to the motor **21** and the switch **27**. In this embodiment, the controller **29** is configured to control driving of the motor **21** (energization to the motor **21**) according to ON/OFF of the switch **27**.

(73) A battery mounting part **18** is provided in the controller housing part **17**. A rechargeable battery **93** is removably mounted to the battery mounting part **18**. The battery mounting part **18** includes a structure for physical engagement with the battery **93** and terminals for electrical

connection with the battery **93**. The grinder **1A** of this embodiment operates by power supplied from the battery **93**. The grinder **1A** may however be configured to operate by power supplied from an external AC power source via a power cord.

(74) The cover **5A** is now described. The cover **5A** is removable from the housing **10A**, but in the following description, the directions of the cover **5A** are referred to based on the directions of the grinder **1A** with the cover **5A** mounted to the housing **10A**.

(75) As shown in FIGS. **1**, **3** and **4**, the cover **5A** includes a cover body **50** and a rail **55A**. The cover body **80** is a single (seamless) metal member. The rail **55A** is a metal member that is originally (initially) formed separately (discretely, individually) from the cover body **50** and thereafter fixed to the cover body **50**. Owing to this structure, manufacture of the cover **5A** can be facilitated, because the cover body **50** and the rail **55A** are joined together after being formed separately. The cover body **50** and the rail **55A** may however be integrally formed as a single member from the beginning.

(76) The cover body **50A** is configured to partially cover the tool accessory **91**. The cover body **50A** of this embodiment includes an upper plate part **51**, an outer peripheral (circumferential) part **53** and a lip part **54**.

(77) The upper plate part **51** is a thin plate-like part that is to be arranged above the tool accessory **91** in the up-down direction. The upper plate part **51** is arranged substantially in parallel to a plane that is orthogonal to the drive axis DX (i.e., substantially in parallel to a plane of the disc-like tool accessory **91** such as a grinding wheel, a cutting wheel and a blade). The upper plate part **51** has a fan-like (arcuate) shape as viewed from above or below, and has an arcuate inner edge (radially inner edge) **511** and an arcuate outer edge (radially outer edge) **515**. The upper plate part **51** is configured to cover a portion of the disc-like tool accessory **91** by the center angle of 180 degrees or more.

(78) The outer peripheral part **53** is formed along the outer edge **515** of the upper plate part **51** and protrudes downward from the upper plate part **51**. The outer peripheral part **53** has an arcuately curved thin plate-like shape. The outer peripheral part **53** is to be arranged radially outward of the outer edge of the disc-like tool accessory **91**. The outer peripheral part **53** is configured such that its lower end is located at least below a lower surface of the disc-like tool accessory **91** in the up-down direction.

(79) The lip part **54** protrudes radially inward from a lower end of the outer peripheral part **53**. An inner edge of the lip part **54** is located radially outward of the outer edge of the disc-like tool accessory **91**. The width of the lip part **54** in the radial direction may be changed from the example of this embodiment, or the lip part **54** may be omitted.

(80) The rail **55A** is a thin plate-like part that protrudes radially outward from the cover body **50**. More specifically, the rail **55A** protrudes radially outward from an upper end of the outer peripheral part **53**. The rail **55A** is positioned such that an upper surface **551** of the rail **55A** is substantially flush with (substantially at the same location in the up-down direction as) an upper surface **513** of the upper plate part **51**. The width (radial length) of the rail **55A** is substantially constant and is much smaller than the radial length of the upper plate part **51**. The thickness of the rail **55A** in the up-down direction is substantially constant. With such a structure, the rail **55A** may also be said as an arcuate protruding part that protrudes radially outward of the upper plate part **51** or the outer peripheral part **53**.

(81) The rail **55A** has multiple engagement holes **56A** that are spaced apart from each other. In this embodiment, all the engagement holes **56A** are at equal intervals, but the engagement holes **56A** may be arranged at different intervals. Each of the engagement holes **56A** is formed through the rail **55A** in the up-down direction (i.e., in the thickness direction of the rail **55A**). The engagement hole **56A** may however be a recess (bottomed hole) that is open downward, or a recess (notch) recessed radially inward from an outer edge of the rail **55A**. The engagement hole **56A** of this embodiment has a circular section.

(82) A structure that is provided on/in the housing **10A** for mounting the cover **5A** to the housing **10A** is now described.

(83) As shown in FIGS. **1**, **3**, **5** and **6**, the housing **10A** is provided with a body engagement part **31A** that is configured to engage with the cover body **50** (specifically, the upper plate part **51**) of the cover **5A**, and a rail engagement part **30A** that is configured to engage with the rail **55A**.

(84) As shown in FIGS. **3**, **5** and **6**, the body engagement part **31A** includes a body engagement groove **311**. The body engagement groove **311** is an annular groove formed in the housing **10A** such that the body engagement groove **311** surrounds (encircles) the drive axis **DX** (the spindle **25**). The body engagement groove **311** is configured to receive at least a portion of an inner edge part **512** (an arcuate part along the inner edge **511**) of the upper plate part **51** of the cover body **50**. More specifically, the body engagement groove **311** and the inner edge part **512** are configured to have generally complementary sectional shapes. When the user mounts the cover **5A** to the housing **10A**, the user first places the cover **5A** such that the upper plate part **51** is substantially orthogonal to the drive axis **DX** and the inner edge part **512** faces the body engagement groove **311** from the front of the housing **10A**. Then the user moves the cover **5A** rearward relative to the housing **10A** and fits the inner edge part **512** into the body engagement groove **311**. Further, the user can turn the cover **5A** around the drive axis **DX** relative to the housing **10A** with the inner edge part **512** of the upper plate part **51** fitted in the body engagement groove **311**.

(85) The rail engagement part **30A** includes a rail engagement groove **32A** and a lever **35A**.

(86) As shown in FIGS. **3**, **5** and **6**, the rail engagement groove **32A** is radially outward of the cover **5A** in the housing **10A**. In this embodiment, the rail engagement groove **32A** is formed substantially in the center of the housing **10A** in the left-right direction behind the body engagement groove **311**. The rail engagement groove **32A** is open forward and is recessed rearward, and extends arcuately. The rail engagement groove **32A** is configured to receive a portion of an outer edge part **556** (an arcuate part extending along the outer edge **555**) of the rail **55A**. The rail engagement groove **32A** and the outer edge part **556** are configured to have generally complementary sectional shapes. As described above, the upper surface **551** of the rail **55A** is substantially flush with the upper surface **513** of the upper plate part **51** of the cover body **50**, so that upper ends of the body engagement groove **311** and the rail engagement groove **32A** are located substantially at the same location in the up-down direction.

(87) The rail engagement groove **32A** faces a portion of the body engagement groove **311** that is located behind the spindle **25** in the front-rear direction. The rail engagement groove **32A** is configured such that the distance between a bottom surface (depth end surface) of the body engagement groove **311** and a bottom surface of the rail engagement groove **32A** is slightly longer than the distance between the inner edge **511** of the upper plate part **51** and the outer edge **555** of the rail **55A** in the radial direction. Owing to this structure, when the user turns the cover **5A** with the inner edge part **512** of the upper plate part **51** fitted in the body engagement groove **311** as described above, the outer edge **555** of the rail **55A** is fitted into the rail engagement groove **32A**. In this state, the user can turn the cover **5A** to a desired position. The cover **5A** is rotated while the inner edge part **512** of the upper plate part **51** and the outer edge part **556** of the rail **55A** slide in the body engagement groove **311** and the rail engagement groove **32A**, respectively.

(88) As shown in FIGS. **1**, **5** and **6**, the lever **35A** is provided on/in the housing **10A** and located radially outward of the cover **5A**. In this embodiment, the lever **35A** is disposed on a left rear end portion of the driving-mechanism housing part **11**. The lever **35A** is supported to be linearly movable. The lever **35A** is partially exposed to the outside of the housing **10A** and partially disposed within the housing **10A**.

(89) More specifically, as shown in FIGS. **1** and **5** to **7**, the lever **35A** includes a manipulation part **351A** that is configured to be manipulated by the user, an engagement part **353A** that is configured to engage with the rail **55A**, and a connection part **355A** that connects the manipulation part **351A** and the engagement part **353A**. In this embodiment, the manipulation part **351A**, the connection

part 355A and the engagement part 353A are formed as a single (seamless) metal member. The lever 35A may, however, be formed by fixedly connecting originally separate members together.

(90) The connection part 355A is a plate-like part that linearly extends in the up-down direction. The manipulation part 351A protrudes from one end (upper end) of the connection part 355A in its longitudinal direction, and extends in a direction (forward) that intersects the connection part 355A. The engagement part 353A protrudes from the other end (lower end) of the connection part 355A in the longitudinal direction, and extends in the direction (forward) crossing the connection part 355A. An engagement projection 354A protrudes from a tip end (distal end, leading end) of the engagement part 353A toward the manipulation part 351A (upward). The engagement projection 354A is configured to engage with any one of the engagement holes 56A of the rail 55A of the cover 5A. The width of the tip end of the engagement projection 354A in the left-right direction is set slightly smaller than the diameter of the engagement hole 56A.

(91) The lever 35A is supported such that the manipulation part 351A is exposed upward from the housing 10A, the connection part 355A is at least partially within the housing 10A, the engagement part 353A is exposed downward from the housing 10A, and the engagement projection 354A protrudes upward.

(92) More specifically, as shown in FIGS. 6 and 7, a portion of the connection part 355A is disposed within a holding groove 101A that is formed in a rear end portion of the driving-mechanism housing part 11. The portion of the connection part 355A is slidable substantially in parallel to the drive axis DX (i.e., substantially in the up-down direction). The holding groove 101A is recessed to the right from a left side surface of the housing 10A (the driving-mechanism housing part 11) and has open upper and lower ends. Upper and lower ends of the connection part 355A protrude upward and downward from the holding groove 101A, respectively. The manipulation part 351A protrudes forward from the connection part 355A above the holding groove 101A. The engagement part 353A protrudes forward from the connection part 355A below the holding groove 101A. The holding groove 101A is covered by a cover part 102A from the left. In this manner, the holding groove 101A and the cover part 102A are utilized to house a portion (the connection part 355A) of the lever 35A within the housing 10A. This configuration facilitates assembling. Further, this configuration can reduce a possibility that the lever 35A is affected by an unintended (unexpected) external force.

(93) In this embodiment, the lever 35A is biased upward by a biasing member 357A. More specifically, a spring receiving projection 356A protrudes rearward from the connection part 355A of the lever 35A. A spring receiving recess 103A is formed on the rear side of the holding groove 101A of the housing 10A and communicates with the holding groove 101A. The spring receiving projection 356A is arranged within the spring receiving recess 103A. The biasing member 357A is a compression coil spring. The biasing spring 357A is below the spring receiving projection 356A within the spring receiving recess 103A. Therefore, in an initial state in which an external force is not applied against the biasing force of the biasing member 357A, the lever 35A is held in an uppermost position where the spring receiving projection 356A abuts a surface that defines an upper end of the spring receiving recess 103A.

(94) When the lever 35A is at the uppermost position, the manipulation part 351A of the lever 35A is spaced apart upward from an upper end of the holding groove 101A. The engagement projection 354A of the engagement part 353A is at a position where the engagement projection 354A is engageable with the engagement hole 56A of the rail 55A of the cover 5A. More specifically, when the engagement projection 354A is at a position corresponding to any one of the engagement holes 56A of the rail 55A in the circumferential direction around the drive axis DX, and the lever 35A is at the uppermost position, as shown in FIG. 8, the engagement projection 354A protrudes into the engagement hole 56A from below and engages with the engagement hole 56A. Further, engagement between the engagement projection 354A and the engagement hole 56A is stably maintained by the biasing force of the biasing member 357A. The position of the lever 35A at

which the lever **35A** is engageable with the rail **55A** (specifically, the position at which the engagement projection **354A** is engageable with any one of the engagement holes **56A**) is hereinafter also referred to as an engagement position of the lever **35A**.

(95) When the lever **35A** is at the engagement position and the engagement projection **354A** is engaged with one of the engagement holes **56A**, the engagement projection **354A** and the engagement hole **56A** can reliably restrict rotation of the cover **5A** relative to the housing **10A**. Further, engagement between the rail **55A** (the outer edge part **556**) and the rail engagement groove **32A** can restrict movement of the cover body **50** in the up-down direction relative to the housing **10A**. Similarly, since the upper plate part **51** (the inner edge part **512**) of the cover body **50** of the cover **5A** is also engaged with the body engagement groove **311**, movement of the cover **5A** in the up-down direction can be restricted at two locations that are spaced apart from each other in the front-rear direction. Thus, movement of the cover body **50** in the up-down direction relative to the housing **10A** can be effectively restricted. Further, movement of the cover body **50** in the front-rear direction relative to the housing **10A** can also be effectively restricted.

(96) The lever **35A** can be moved downward from the engagement position in response to user's manipulation. Specifically, when the user pushes the manipulation part **351A** downward, as shown in FIGS. **9** and **10**, the lever **35A** is moved downward against the biasing force of the biasing member **357A** and the engagement projection **354A** of the engagement part **353A** is disengaged from the engagement hole **56A** of the rail **55A**. Thus, the engagement between the engagement projection **354A** and the engagement hole **56A** is released. The position of the lever **35A** at which the lever **35A** is not engageable with the rail **55A** (specifically, the position at which the engagement projection **354A** cannot engage with the engagement hole **56A**) is hereinafter also referred to as a disengagement position of the lever **35A**. When the lever **35A** is at the disengagement position, the user can turn the cover **5A** relative to the housing **10A** around the drive axis **DX** to a desired position, or remove the cover **5A** from the housing **10A**.

(97) In this embodiment, the manipulation part **351A** of the lever **35A** is exposed upward from the housing **10A**. Therefore, the user can easily move the lever **35A** from the engagement position to the disengagement position. Further, the lever **35A** is configured to be linearly moved when the user pushes the manipulation part **351A** downward. Therefore, the user can manipulate the manipulation part **351A** more easily, compared with a structure that requires the user to pull the manipulation part **351A** upward.

(98) As described above, in the grinder **1A** of this embodiment, the rail **55A** provided on the cover body **50** can restrict (i) movement of the cover body **50** in the circumferential direction (i.e., rotation around the drive axis **DX**) relative to the housing **10A**, and (ii) movement of the cover body **50** in other directions (specifically, the up-down direction and the front-rear direction) relative to the housing **10A**, in cooperation with the rail engagement part **30A** (the rail engagement groove **32A** and the lever **35A**) provided on/in the housing **10A**. Thus, the rail **55A** performs two functions of (i) properly positioning the cover body **50** in the circumferential direction relative to the housing **10A**, and (ii) suppressing looseness or play in the other directions. Therefore, compared with a case in which two separate (discrete, individual) structures respectively perform these two functions, a space required in the power tool can be smaller and the power tool can be simplified in structure. In this embodiment, the upper surface **551** of the rail **55A** is flush with the upper surface **513** of the upper plate part **51**, so that the upper surfaces **551**, **513** can be integrally utilized to restrict upward movement of the cover body **50** (i.e., to suppress looseness in the up-down direction).

(99) Further, the rail **55A** is a protruding part protruding radially outward from the cover body **50** and not protruding upward or downward from the cover body **50**. In addition, the rail engagement groove **32A** of the rail engagement part **30A** is above the lower end of the cover body **50**, and the lower end of the lever **35A** is above the lower end of the cover body **50** both when the lever **35A** is at the engagement position and at the disengagement position. This configuration can suppress size increase of the entire grinder **1A** (particularly, a front end portion of the grinder **1A**) in the up-down

direction. Further, the rail 55A and the rail engagement part 30A do not interfere with (get in the way of) the processing operation. Thus, this embodiment provides the grinder 1A that is easy to operate even in a relatively narrow space.

Second Embodiment

(100) A grinder 1B of a second embodiment is now described with reference to FIG. 11. The grinder 1B of the second embodiment has substantially the same structure (including a slight difference in shape) as that of the grinder 1A of the first embodiment except for a housing 10B. Most of the housing 10B has substantially the same structure as that of the grinder 1A of the first embodiment. Therefore, components of the housing 10B that are substantially identical to those of the grinder 1A are given the same numerals and their description is omitted or simplified, and different structures are mainly described. The same applies to subsequent embodiments.

(101) As shown in FIG. 11, the housing 10B of the second embodiment has the same body engagement part 31A as that of the first embodiment and a rail engagement part 30B that is different from the rail engagement part 30A of the first embodiment. In the other points, the housing 10 has substantially the same structure as the housing 10A of the first embodiment. The rail engagement part 30B includes the same rail engagement groove 32A (see FIG. 3) as that of the first embodiment and a lever 35B that is different from the lever 35A (see FIG. 7) of the first embodiment.

(102) The lever 35B is configured such that the operating direction is reverse to that of the lever 35A. More specifically, the lever 35B includes the manipulation part 351A that is configured to be manipulated by a user, an engagement part 353B that is configured to engage with any one of the engagement holes 56A of the rail 55A, and the connection part 355A that connects the manipulation part 351A and the engagement part 353B. Like in the first embodiment, a portion of the connection part 355A is held to be slidable in the up-down direction within the holding groove 101A. The manipulation part 351A protrudes forward from the upper end of the connection part 355A above the holding groove 101A. The engagement part 353B protrudes forward from the lower end of the connection part 355A. An engagement projection 354B protrudes downward from a tip end (distal end) of the engagement part 353B. Thus, the engagement projection 354B protrudes in the opposite direction to the direction in which the engagement projection 354A of the first embodiment protrudes. The width of the tip end of the engagement projection 354B in the left-right direction is set slightly smaller than the diameter of the engagement hole 56A.

(103) In this embodiment, the biasing member 357A is disposed on the upper side of the spring receiving projection 356A within the spring receiving recess 103A. Thus, the lever 35B is biased downward by the biasing member 357A. In an initial state, the lever 35B is held in a lowermost position where the spring receiving projection 356A abuts a surface that defines a lower end of the spring receiving recess 103A. When the lever 35B is at the lowermost position, the engagement projection 354B is engageable with one of the engagement holes 56A of the rail 55A. Thus, in this embodiment, the lowermost position of the lever 35B corresponds to the engagement position of the lever 35B, and the disengagement position of the lever 35B is defined above the engagement position. Although not shown, the user can move the lever 35B upward to the disengagement position by pulling up the manipulation part 351A against the biasing force of the biasing member 357A.

(104) In this embodiment, the rail 55A and the rail engagement part 30B also perform two functions of (i) properly positioning the cover body 50 in the circumferential direction relative to the housing 10B and (ii) suppressing looseness in other directions, in cooperation with each other. Therefore, compared with a case in which separate structures respectively perform these two functions, a space required in the power tool can be reduced and the power tool can be simplified in structure.

(105) Further, the rail 55A is a protruding part protruding radially outward from the cover body 50 and not protruding upward or downward from the cover body 50. The lever 35B is above the lower

end of the cover body **50** both in the engagement position and in the disengagement position. Therefore, in this embodiment, size increase of the entire grinder **1B** (particularly, a front end portion of the grinder **1B**) can also be suppressed in the up-down direction. Thus, this embodiment provides the grinder **1B** that is easy to operate even in a relatively narrow space.

Third Embodiment

(106) A grinder **1C** of a third embodiment is now described with reference to FIGS. **12** to **16**. The grinder **1C** includes a cover **5C** and a housing **10C** that are partially different in structure from the cover **5A** and the housing **10A** of the first embodiment.

(107) As shown in FIGS. **12** and **13**, the cover **5C** includes the cover body **50** and a rail **55C**. The rail **55C** has substantially the same structure as the rail **55A** (see FIG. **4**) of the first embodiment. The rail **55C** is an arcuate protruding part protruding radially outward from the cover body **50** and has the engagement holes **56A**. Unlike in the first embodiment, however, the rail **55C** protrudes substantially from the center of the outer peripheral part **53** in the up-down direction.

(108) As shown in FIGS. **12**, **14** and **15**, the housing **10C** has the same body engagement part **31A** (the body engagement groove **311**) as that of the first embodiment and a rail engagement part **30C** that is different from the rail engagement part **30A** of the first embodiment. In the other points, the housing **10C** has substantially the same structure as the housing **10A** of the first embodiment.

(109) The rail engagement part **30C** is provided behind the cover **5C** and includes a lever **35C** that is different from the lever **35A** of the first embodiment. The lever **35C** is partially housed in a lower portion of the housing **10C**. The lever **35C** is configured to be manipulated from below the housing **10C** by a user. More specifically, the lever **35C** includes a manipulation part **351C** that is configured to be manipulated by the user, an engagement part **353C** that is configured to engage with any one of the engagement holes **56A** of the rail **55C**, and a connection part **355C** that connects the manipulation part **351C** and the engagement part **353C**.

(110) The connection part **355C** is a plate-like part that extends linearly. The connection part **355C** is arranged to be slidable in the up-down direction within a holding groove **101C** that is formed in a lower end portion of the housing **10C**. The holding groove **101C** is substantially in the center of the housing **10C** in the left-right direction. The manipulation part **351C** is connected to a lower end of the connection part **355C** below the holding groove **101C** so as to form a T-shape together with the connection part **355C**. The engagement part **353C** protrudes from an upper end of the connection part **355C** forward of the holding groove **101C**. An engagement projection **354C** is formed on a tip end of the engagement part **353C** and protrudes downward. The width of the tip end of the engagement projection **354C** in the left-right direction is set slightly smaller than the diameter of the engagement hole **56A**.

(111) The lever **35C** is biased downward by a biasing member **357C**. The biasing member **357C** is within a spring receiving recess **103C** formed in the housing **10C**. The spring receiving recess **103C** is configured to communicate with the holding groove **101C** on the front side of the holding groove **101C** and to be recessed upward. The engagement part **353C** of the lever **35C** is within the spring receiving recess **103C**. The spring receiving recess **103C** is covered by a cover part **102C** from below. In this manner, the spring receiving recess **103C** and the cover part **102C** are utilized to arrange a portion of the lever **35C** within the housing **10C**. This configuration facilitates assembling. Further, this configuration can reduce the possibility that the lever **35C** is affected by an unintended external force.

(112) The biasing member **357C** is a compression coil spring and is disposed above the engagement part **353C** within the spring receiving recess **103C**. Therefore, as shown in FIG. **12**, in the initial state, the lever **35C** is held in a lowermost position where the engagement part **353C** abuts (is in contact with) an upper surface of the cover part **102C**. When the lever **35C** is at the lowermost position, the engagement projection **354C** is engageable with any one of the engagement holes **56A** of the rail **55C**. Thus, in this embodiment, the lowermost position of the lever **35C** corresponds to the engagement position of the lever **35C**, and the disengagement position

of the lever 35C is defined above the engagement position. As shown in FIG. 16, the user can move the lever 35C upward to the disengagement position by pushing the manipulation part 351C upward against the biasing force of the biasing member 357C.

(113) As described above, in the grinder 1C of this embodiment, engagement between the rail 55C provided on the cover body 50 and the lever 35C provided on/in the housing 10C can restrict rotation of the cover body 50 relative to the housing 10C. In this embodiment, engagement between the engagement projection 354C of the lever 35C and the engagement hole 56A of the rail 55C can also reliably restrict rotation of the cover body 50 relative to the housing 10C.

(114) The rail 55C is a protruding part protruding radially outward from the cover body 50 and not protruding upward or downward from the cover body 50. In this embodiment, an entirety of the lever 35C is between the upper end and the lower end of the cover body 50 both in the engagement position and in the disengagement position. This configuration can also suppress size increase of the entire grinder 1C (particularly, a front end portion of the grinder 1C) in the up-down direction. Further, the rail 55C and the lever 35C do not interfere with the processing operation. Thus, this embodiment provides the grinder 1C that is easy to operate even in a relatively narrow space.

Fourth Embodiment

(115) A grinder 1D of a fourth embodiment is now described with reference to FIGS. 17 to 21. The grinder 1D includes a cover 5D and a housing 10D that are partially different in structure from the cover 5A and the housing 10A of the first embodiment.

(116) As shown in FIGS. 17 and 18, the cover 5D includes the cover body 50 and a rail 55D. Like the rail 55A (see FIG. 4) of the first embodiment, the rail 55D is configured as an arcuate protruding part protruding radially outward. Unlike the rail 55A, however, the rail 55D protrudes substantially from the center of the outer peripheral part 53 in the up-down direction. The rail 55D has engagement recesses 56D formed in the outer edge. Each of the engagement recesses 56D is a rectangular recess (notch) recessed radially inward from the outer edge of the rail 55D. In this embodiment, all the engagement recesses 56D are at equal intervals. Instead, the engagement recesses 56D may be at different intervals.

(117) As shown in FIGS. 17, 19 and 20, the housing 10D has the same body engagement part 31A (the body engagement groove 311) as that of the first embodiment and a rail engagement part 30D that is different from the rail engagement part 30A of the first embodiment. In the other points, the housing 10D has substantially the same structure as the housing 10A of the first embodiment.

(118) The rail engagement part 30D is behind the cover 5D and includes a lever 35D that is different from the lever 35A of the first embodiment. The lever 35D is a rotary lever. The lever 35 is supported behind the cover 5D on/in a lower end portion of the housing 10D. The lever 35D is configured to be manipulated on a lateral side of the housing 10D by a user. More specifically, the lever 35D includes a manipulation part 351D that is configured to be manipulated by the user, an engagement part 353D that is configured to engage with any one of the engagement recesses 56D of the rail 55D, and a connection part 355D that connects the manipulation part 351D and the engagement part 353D.

(119) The connection part 355D is a plate-like part that extends linearly. The connection part 355D is pivotably (rotatably) mounted via a screw 358D to a cylindrical part 105 provided in the lower end portion of the housing 10D. The connection part 355D linearly extends substantially in the left-right direction. The manipulation part 351D is connected to a left end of the connection part 355D and located leftward of the housing 10D. The manipulation part 351D is L-shaped and includes a tab, which protrudes leftward. The engagement part 353D includes an engagement projection 354D that protrudes downward from a right end of the connection part 355D. The engagement projection 354D is substantially in the center of the housing 10D in the left-right direction. The width of the tip end of the engagement projection 354D in the left-right direction is set slightly smaller than the width of the engagement recess 56D in the circumferential direction.

(120) In this embodiment, the lever 35D is pivotally biased by a biasing member 357D to be

pivoted (turned, rotated) in such a direction that the engagement projection **354D** is moved forward (i.e., toward the cover **5D**). More specifically, the biasing member **357D** is a torsion coil spring. A coil part of the biasing member **357D** is fitted around the cylindrical part **105**, into which the screw **358D** is threadedly engaged. A spring receiving projection **356D** protrudes upward between the screw **358D** and the engagement projection **354D** in the connection part **355D** of the lever **35D**. One end of the biasing member **357D** is locked to the housing **10D**, and the other end of the biasing member **357D** is locked to the spring receiving projection **356D**. Thus, the lever **35D** is pivotally biased around an axis extending in the up-down direction such that the engagement projection **354D** is moved forward.

(121) When the engagement projection **354D** is at a position corresponding to any one of the engagement recesses **56D** in the circumferential direction, and the lever **35D** is at the frontmost position, as shown in FIG. **19**, the engagement projection **354D** is inserted into the engagement recess **56D** from radially outside. In this embodiment, the position of the lever **35D** in which the engagement projection **354D** is in the frontmost position corresponds to the engagement position of the lever **35D**. As shown in FIG. **21**, when the user pushes the manipulation part **351D** (tab) forward, the lever **35D** is pivoted (turned, rotated) around the axis extending in the up-down direction against the biasing force of the biasing member **357D** in such a direction that the engagement projection **354D** is moved rearward (i.e., away from the cover **5D**). Thus, the lever **35D** is moved to the disengagement position at which the engagement projection **354D** is disengaged from the engagement recess **56D** of the rail **55D**.

(122) As described above, in the grinder **1D** of this embodiment, engagement between the rail **55D** provided on the cover body **50** and the lever **35D** provided on/in the housing **10D** can restrict rotation of the cover body **50** relative to the housing **10D**. In this embodiment, engagement between the engagement projection **354D** of the lever **35D** and the engagement recess (notch) **56D** of the rail **55D** can also reliably restrict rotation of the cover body **50** relative to the housing **10D**.

(123) The rail **55D** is a protruding part protruding radially outward from the cover body **50** and not protruding upward or downward from the cover body **50**. In this embodiment, the lever **35D** is a rotary (pivotable) lever that is movable within (along) a plane that is substantially orthogonal to the drive axis **DX**. An entirety of the lever **35D** is located between the upper end and the lower end of the cover body **50** both in the engagement position and in the disengagement position. This configuration can also suppress size increase of the entire grinder **1D** (particularly, a front end portion of the grinder **1D**) in the up-down direction. Further, the rail **55D** and the lever **35D** do not interfere with the processing operation. Thus, this embodiment provides the grinder **1D** that is easy to operate even in a relatively narrow space.

Fifth Embodiment

(124) A grinder **1E** of a fifth embodiment is now described with reference to FIGS. **22** to **28**. The grinder **1E** includes a cover **5E** and a housing **10E** that are partially different in structure from the cover **5A** and the housing **10A** of the first embodiment.

(125) As shown in FIGS. **22** and **23**, the cover **5E** includes the cover body **50** and a rail **55E**. Like the rail **55A** of the first embodiment, the rail **55E** is configured as an arcuate protruding part protruding radially outward. The upper surface **551** of the rail **55E** is substantially flush with the upper surface **513** of the upper plate part **51**. Unlike in the first embodiment, however, the rail **55E** has engagement recesses **56E** formed in the upper surface **551**. Each of the engagement recesses **56E** is a hemispherical recess recessed downward from the upper surface **551**. In this embodiment, all the engagement recesses **56E** are at equal intervals. Instead, the engagement recesses **56E** may be at different intervals.

(126) As shown in FIGS. **22** and **24** to **26**, the housing **10E** has the same body engagement part **31A** (the body engagement groove **311**) as that of the first embodiment and a rail engagement part **30E** that is different from the rail engagement part **30A** of the first embodiment. In the other points, the housing **10E** has substantially the same structure as the housing **10A** of the first embodiment.

(127) The rail engagement part **30E** is behind the cover **5E**. The rail engagement part **30E** includes an engagement projection **32E** (see FIGS. **24** and **26**), a restriction wall **33** and a lever **35E**.

(128) The engagement projection **32E** is directly above the rail **55E** of the cover **5E**. The engagement projection **32E** protrudes downward from a portion of a lower surface **104** that is in a central portion of the housing **10E** in the left-right direction. The engagement projection **32E** has a shape that is generally complementary to the shape of the engagement recess **56E** of the rail **55E**. Thus, the engagement projection **32E** has a hemispherical shape.

(129) The restriction wall **33** is a wall part extending in the up-down direction behind the engagement projection **32E**. A front surface of the restriction wall **33** is gently curved, corresponding to the outer edge **555** of the rail **55E**. The restriction wall **33** is configured such that the distance between the bottom surface (depth end surface) of the body engagement groove **311** and a front surface of the restriction wall **33** is slightly longer than the distance between the inner edge **511** of the upper plate part **51** and the outer edge **555** of the rail **55E** in the radial direction.

(130) The lever **35E** is a rotary (pivotal) lever and is supported on a lower end portion of the housing **10E** behind the cover **5E**. The lever **35E** is configured to be manipulated under the housing **10E** by a user. More specifically, the lever **35E** is generally L-shaped as viewed from the side (from the left or the right). The lever **35E** includes a manipulation part **351E** that is configured to be manipulated by the user, an engagement part **353E** that is configured to engage with (abut) the rail **55E**, and a connection part **355E** that connects the manipulation part **351E** and the engagement part **353E**.

(131) A pair of left and right arm parts **356E** protrude from the connection part **355E**. The connection part **355E** is pivotably (rotatably) mounted to a lower end portion of the housing **10E** via a support pin **358E**. The support pin **358E** is inserted through the arm parts **356E** and supported by a pair of left and right support parts **107** of the housing **10E**. The support pin **358E** extends in the left-right direction. Thus, the lever **35E** is pivotable (turnable, rotatable) around an axis extending in the left-right direction. The manipulation part **351E** extends substantially rearward from one end of the connection part **355E** under the housing **10E**. The engagement part **353E** extends substantially upward from the other end of the connection part **355E**. The engagement part **353E** of this embodiment is configured to engage with (abut) the rail **55E** and press the rail **55E** against the housing **10E** (the lower surface **104**).

(132) In this embodiment, the lever **35E** is pivotally biased by a biasing member **357E** to be pivoted (turned, rotated) in such a direction that the engagement part **353E** is moved substantially upward (i.e. toward the rail **55E** from below). More specifically, the biasing member **357E** is a torsion coil spring. A coil part of the biasing member **357E** is fitted around the support pin **358E**. One end of the biasing member **357E** is locked to the housing **10E**, and the other end of the biasing member **357E** is locked to the connection part **355E**. Thus, the lever **35E** is biased to be pivoted (turned) around an axis extending in the left-right direction such that the engagement part **353E** is moved substantially upward.

(133) As shown in FIG. **22**, when the engagement part **353E** is at its uppermost position, a tip end of the engagement part **353E** abuts (is held in contact with) a lower surface **552** of the rail **55E** of the cover **5E** and presses the rail **55E** against the lower surface **104** of the housing **10E**. When the engagement projection **32E** on the lower surface of the housing **10E** is at a position corresponding to any one of the engagement recesses **56E** of the cover **5E** in the circumferential direction, as shown in FIG. **24**, the engagement projection **32E** engages with the engagement recess **56E**. The position of the lever **35E** at which the engagement part **353E** engages with (abuts) the rail **55E** to press the rail **55E** against the housing **10E** to thereby cause the engagement projection **32E** and the engagement recess **56E** to be engaged with each other is hereinafter also referred to as an engagement position of the lever **35E**. When the lever **35E** is at the engagement position, the manipulation part **351E** extends obliquely rearward and downward.

(134) As shown in FIGS. **27** and **28**, when a user pushes the manipulation part **351E** upward toward

the housing **10E**, the lever **35E** is pivoted (turned, rotated) against the biasing force of the biasing member **357E** such that the engagement part **353E** is moved downward (i.e., away from the rail **5E**). Thus, the engagement projection **32E** is disengaged from the engagement recess **56E**. The position of the lever **35E** in which the engagement projection **32E** is not engageable with the engagement recess **56E** is hereinafter also referred to as a disengagement position of the lever **35E**. When the lever **35E** is at the disengagement position, the user can turn the cover **5E** to a desired position around the drive axis **DX** relative to the housing **10E**, or remove the cover **5E** from the housing **10E**.

(135) In this embodiment, as shown in FIGS. **22** and **27**, when the lever **35E** is both in the engagement position and in the disengagement position, a rear end of the manipulation part **351E** (a tip end of the lever **35E**) is arranged at least forward of the handle part **15** (see FIG. **2**) in the front-rear direction. In this embodiment, the rear end of the manipulation part **351E** is forward of the switch knob **271** supported on the motor housing part **13**. More specifically, the rear end of the manipulation part **351E** is located directly below a front end portion of the motor housing part **13** (more specifically, directly below the fan **22**). With such arrangement, when the user holds the handle part **15** and operates the switch knob **271**, the lever **35E** does not easily interfere with this operation.

(136) As described above, in the grinder **1E** of this embodiment, the rail **55E** and the rail engagement part **30E** cooperate with each other to perform two functions of (i) properly positioning the cover body **50** in the circumferential direction relative to the housing **10E** and (ii) suppressing looseness in the other directions. Therefore, compared with a case in which two separate structures respectively perform these two functions, a space required in the power tool can be reduced and the power tool can be simplified in structure.

(137) Specifically, the lever **35E**, which is biased by the biasing member **357E**, can effectively restrict movement of the rail **55E** relative to the housing **10E** in the up-down direction by pressing the rail **55E** against the lower surface **104** of the housing **10E**. In this embodiment, the upper surface **551** of the rail **55E** is flush with the upper surface **513** of the upper plate part **51**, so that the upper surfaces **551**, **513** are integrally pressed against the lower surface **104** of the housing **10E**, and movement of the cover **5E** in the up-down direction can be more reliably restricted.

(138) Further, in this embodiment, the lever **35E** presses the rail **55E** against the lower surface **104** of the housing **10E** with the hemispherical engagement projection **32E** of the housing **10E** being engaged with the hemispherical engagement recess **56E** of the rail **55E**. This configuration can effectively suppress movement of the cover **5E** in all directions (e.g., the circumferential direction, the front-rear direction and the up-down direction) relative to the housing **10E**. It is noted, however, that the engagement projection **32E** may be omitted. Even in such a modification, movement of the cover **5E** relative to the housing **10E** can be restrained not only in the up-down direction but also in other directions (e.g., the circumferential direction and the front-rear direction) by the lever **35E** pressing the rail **55E** against the lower surface **104** of the housing **10E**. Further, the restriction wall **33** can effectively restrict movement of the cover **5E** in the front-rear direction.

Sixth Embodiment

(139) A grinder **1F** of a sixth embodiment is now described with reference to FIGS. **29** to **34**. The grinder **1F** includes a cover **5F** and a housing **10F** that are partially different in structure from the cover **5A** and the housing **10A** of the first embodiment.

(140) As shown in FIGS. **29** and **30**, the cover **5F** includes the cover body **50** and a rail **55F**. Like the rail **55A** (see FIG. **4**) of the first embodiment, the rail **55F** is configured as an arcuate protruding part protruding radially outward. Unlike in the first embodiment, however, the upper surface **551** of the rail **55F** is below the upper surface **513** of the upper plate part **51** of the cover body **50** in the up-down direction. The rail **55F** has engagement recesses **56F** that are similar to the engagement recesses **56A** of the first embodiment. In this embodiment, all the engagement recesses **56F** are at equal intervals. Instead, the engagement recesses **56F** may be at different intervals.

(141) As shown in FIGS. 29 to 32, the housing 10F has the same body engagement part 31A (the body engagement groove 311) as that of the first embodiment and a rail engagement part 30F. In the other points, the housing 10E has substantially the same structure as the housing 10A of the first embodiment.

(142) The rail engagement part 30F is behind the cover 5F and includes a rail engagement groove 32F and a lever 35F.

(143) The rail engagement groove 32F is formed substantially in the center of the housing 10A in the left-right direction behind the body engagement groove 311. The rail engagement groove 32F is located below the body engagement groove 311, corresponding to the rail 55F that is arranged below the upper plate part 51. The rail engagement groove 32F has substantially the same structure as the rail engagement groove 32A (see FIG. 3) of the first embodiment. Specifically, the rail engagement groove 32F is open forward and is recessed rearward and extends arcuately, and is configured to receive a portion of the outer edge part 556 of the rail 55F. The rail engagement groove 32F and the outer edge part 556 are configured to have respective sectional shapes that are substantially complementary to each other.

(144) The lever 35F is different from the lever 35E (see FIG. 26) of the fifth embodiment in that the lever 35F has an engagement part 353F in place of the engagement part 353E of the lever 35E. In the other points, the lever 35F has substantially the same structure as the lever 35E. The lever 35F is generally L-shaped as viewed from the side and includes the manipulation part 351E, an engagement part 353F and the connection part 355E that connects the manipulation part 351E and the engagement part 353F.

(145) Like the lever 35E, the lever 35F is behind the cover 5F. The lever 35F is supported via the support pin 358E on/in a lower end portion of the housing 10F to be pivotable around an axis that extends in the left-right direction. In this embodiment, the lever 35F is disposed leftward of the rail engagement groove 32F. The engagement part 353F has an engagement projection 354F on its tip end. The engagement projection 354F is configured to engage with any one of the engagement holes 56F of the rail 55F. The width of the tip end of the engagement projection 354F in the left-right direction is set slightly smaller than the diameter of the engagement hole 56F. Like in the fifth embodiment, the lever 35F is biased by the biasing member 357E to be pivoted in such a direction that the engagement projection 354F of the engagement part 353F is moved substantially upward (i.e., toward the rail 55F from below).

(146) As shown in FIGS. 30 and 32, when the engagement projection 354F is at a position corresponding to any one of the engagement recesses 56F in the circumferential direction, and the engagement projection 354F is placed in the uppermost position, the engagement projection 354F is inserted into the engagement recess 56F from below and engages with the engagement hole 56F. In this embodiment, the position of the lever 35F in which the engagement projection 354F is in the uppermost position corresponds to the engagement position of the lever 35F.

(147) As shown in FIGS. 33 and 34, when a user pushes the manipulation part 351E upward toward the housing 10F, the lever 35F is pivoted around an axis extending in the left-right direction against the biasing force of the biasing member 357E such that the engagement projection 354F is moved downward. Thus, the lever 35F is placed in the disengagement position. In this embodiment, the engagement projection 354F is moved along an arc within the engagement hole 56F when the lever 35F is pivoted. Therefore, the diameter of the engagement hole 56F is set larger than the diameter of the engagement hole 56A of the first embodiment.

(148) Like in the first embodiment, in the grinder 1F of this embodiment, the rail 55F and the rail engagement part 30F (the rail engagement groove 32F and the lever 35F) cooperate with each other to perform two functions of (i) properly positioning the cover body 50 in the circumferential direction relative to the housing 10F and (ii) suppressing looseness in the other directions. Therefore, compared with a case in which separate structures respectively perform these two functions, a space required in the power tool can be reduced and the power tool can be simplified in

structure.

Seventh Embodiment

(149) A grinder **1G** of a seventh embodiment is now described with reference to FIGS. **35** to **37**. The grinder **1G** includes a cover **5G** and a housing **10G** that are partially different in structure from the cover **5A** and the housing **10A** of the first embodiment. The cover **5G** includes only the cover body **50** and does not include a rail. The housing **10G** has the same body engagement part **31A** (the body engagement groove **311**) as that of the first embodiment and an outer peripheral engagement part **30G** that is not provided in the first embodiment. In FIG. **35**, for the sake of convenience, some portions of the housing **10G** of the grinder **10G** and mechanisms or elements disposed within the housing **10G** are simply shown or not shown, but the grinder **10G** has substantially the same structure as the grinder **1A** (see FIG. **3**) of the first embodiment, except for the outer peripheral engagement part **30G**.

(150) As shown in FIGS. **35** to **37**, the outer peripheral engagement part **30G** includes a latch **35G** disposed below the housing **10G**. The latch **35G** is a fastener that utilizes a tensile force. The latch **35G** may also be referred to as a draw latch or a toggle latch. The latch **35G** includes a rectangular plate-like metal engagement part **353G** and a U-shaped metal arm **355G**. One end portion of the engagement part **353G** in its longitudinal direction is bent and forms a hook **354G**. Two opposite ends of the arm **355G** are pivotably (rotatably) connected to opposite sides of the other end portion of the engagement part **353G** in its longitudinal direction. A central portion of the arm **355G** is supported on a lower end portion of the housing **10G** so as to be pivotable around an axis extending in the left-right direction.

(151) Owing to the above-described structure, the latch **35G** is movable between an engagement position at which the hook **354** engages with the lip part **54** as shown in FIGS. **35** and **36**, and a disengagement position at which the hook **354** is not engageable with the lip part **54** as shown in FIG. **37**. When the latch **35G** is in the engagement position, the tensile force is applied to the cover **5G** via the engagement part **353G**, so that movement (rotation around the drive axis **DX**) of the cover **5G** in the circumferential direction relative to the housing **10G** is restricted. The size of the latch **35G** in the up-down direction can be minimized in the engagement position, so that the size of the grinder **1G** in the up-down direction (the extending direction of the drive axis **DX**) can be reduced.

(152) Two other examples of covers that are suitable and effective for use with the latch **35G** are now described with reference to FIGS. **38** and **39**.

(153) A cover **5H** shown in FIG. **38** has projections **541** protruding radially inward from the lip part **54**. The projections **541** are at equal intervals in the circumferential direction. The interval (distance) between the adjacent projections **541** is set slightly larger than the width of the engagement part **353G** of the latch **35G** in the left-right direction. Owing to such a structure, when the latch **35G** is at the engagement position, the hook **354G** (see FIG. **35**) is engaged with the lip part **54** between the projections **541**, so that rotation of the cover **5H** relative to the housing **10G** can be more reliably restricted.

(154) A cover **5J** shown in FIG. **39** does not include the lip part **54** on the outer peripheral part **53**. Instead, the cover **5J** has projections **535** that protrude radially inward from the outer peripheral part **53** (peripheral wall part). The projections **535** are at equal intervals in the circumferential direction. The interval (distance) between the adjacent projections **535** is set slightly larger than the width of the engagement part **353G** of the latch **35G** in the left-right direction. Owing to such a structure, like in the example shown in FIG. **38**, when the latch **35G** is at the engagement position, the hook **354G** engages with a lower edge of the outer peripheral part **53** between the projections **535**, so that rotation of the cover **5J** relative to the housing **10G** can be more reliably restricted.

(155) Correspondences between the features of the above-described embodiments and the features of the present disclosure or invention are as follows. However, the features of the above-described embodiments are merely exemplary and do not limit the features of the present disclosure or

invention.

(156) The grinders **1A**, **1B**, **1C**, **1D**, **1E**, **1F** are examples of a “power tool”. The spindle **25** is an example of a “spindle”. The housings **10A**, **10B**, **10C**, **10D**, **10E**, **10F** are examples of a “housing”. The cover body **50** is an example of a “cover body”. The upper plate part **51** is an example of an “upper plate part”. The outer peripheral part **53** is an example of an “outer peripheral part”. The rails **55A**, **55C**, **55D**, **55E**, **55F** are examples of a “first engagement part” and also a “protruding part”. The levers **35A**, **35B**, **35C**, **35D**, **35E**, **35F** are examples of a “second engagement part”.

(157) The levers **35A**, **35B**, **35C** are examples of a “lever that is linearly movable”. The engagement holes **56A**, **56F** are examples of a “hole of the protruding part”. The lever **35D** is an example of a “rotary lever” that is movable within a plane substantially orthogonal to the drive axis. The engagement recess **56D** is an example of a “recess of the protruding part”. The engagement projections **354A**, **354B**, **354C**, **354D**, **354F** are examples of a “projection of the second engagement part”. The biasing members **357A**, **357C**, **357D**, **357E** are examples of a “biasing member”.

(158) The above-described embodiments are merely exemplary, and the power tool according to the present disclosure is not limited to the grinders **1A**, **1B**, **1C**, **1D**, **1E**, **1F** of the above-described embodiments. For example, the following modifications may be made. At least one of these modifications can be employed in combination with at least one of the features of the grinders **1A**, **1B**, **1C**, **1D**, **1E**, **1F** of the above-described embodiments and the claimed features.

(159) For example, the present disclosure may be applied to other kinds of power tools (e.g., a cutter, a circular saw and a multi-tool) configured such that a cover that is configured to partially covers a tool accessory mounted to a spindle is removably mounted thereto. The structure of the housing, and elements (mechanisms) disposed within the housing and their arrangement can be appropriately changed according to the kind of the power tool.

(160) The cover body according to this disclosure is not limited to the cover body **50** of the above-described embodiments. For example, the shape of the cover body **50** (e.g., the length of the upper plate part **51** in the circumferential direction and/or the radial direction) can be appropriately changed.

(161) The first engagement part according to this disclosure is not limited to the rails **55A**, **55C**, **55D**, **55E**, **55F** of the above-described embodiments. For example, the shapes (e.g., the length in the circumferential direction, the width in the radial direction, and/or the thickness in the up-down direction) of the rails **55A**, **55C**, **55D**, **55E**, **55F** and/or the positions of the rails **55A**, **55C**, **55D**, **55E**, **55F** relative to the cover body **50** can be appropriately changed. The sectional shapes, sizes and/or arrangement of the engagement holes **56A**, **56F** and the engagement recesses **56E** can be appropriately changed. Further, the first engagement part according to this disclosure need not be an arcuate rail, insofar as it is engageable with the second engagement part (e.g., a lever) that is provided in/on the housing so as to be movable between the engagement position and disengagement position. For example, the first engagement part may be configured as a projection, a recess or a hole formed on/in the outer peripheral part of the cover body. The first engagement part according to this disclosure may protrude, not from the outer peripheral part, but from the upper plate part.

(162) The second engagement part according to this disclosure is not limited to the levers **35A**, **35B**, **35C**, **35D**, **35E**, **35F** of the above-described embodiments. For example, the configurations (e.g., the shapes, arrangement, and/or supporting manners) of the levers **35A**, **35B**, **35C**, **35D**, **35E**, **35F** can be appropriately changed in relation to the configurations of the first engagement part (e.g., the recesses or holes of the rails **55A**, **55C**, **55D**, **55E**, **55F**). For example, the second engagement part may be a lever that is linearly movable between the engagement position and the disengagement position within a plane that is orthogonal to the drive axis **DX**. Further, the second engagement part is preferably configured to be biased toward the engagement position by a biasing member, but, for example, it may be configured to be locked to the housing in the engagement

position and in the disengagement position. Springs that are employed as the biasing members 357A, 357C, 357D, 357E in the above-described embodiments are mere examples, and each of them may be changed to another kind of springs, or an elastic element other than a spring.

(163) Further, for example, the sectional shapes and/or arrangement of the rail engagement grooves 32A, 32F can be appropriately changed corresponding to the change of the first engagement part (e.g., the rail). The rail engagement grooves 32A, 32F may be omitted. Alternatively, the housing may have a structure that restricts movement of the cover body relative to the housing in a direction that is different from the circumferential direction, by engaging with the cover body itself, instead of engaging with the first engagement part (e.g., the rail).

(164) In view of the nature of the present disclosure and the above-described embodiments, the following Aspects A1 to A8 are provided. At least one of the following Aspects A1 to A8 can be employed in combination with at least one of the features of the above-described embodiments, its modifications and the claimed features.

(165) (Aspect A1)

(166) The first engagement part is an arcuate rail.

(167) (Aspect A2)

(168) The housing has a rail engagement groove in which an outer edge part of the rail can be fitted.

(169) (Aspect A3)

(170) The second engagement part is at least partially within the housing.

(171) (Aspect A4)

(172) The lever includes: a manipulation part that is configured to be externally manipulated by a user, an engagement part that is configured to engage with the first engagement part when the lever is at the engagement position and not to be engageable with the first engagement part when the lever is at the disengagement position, and a connection part that connects the manipulation part and the engagement part.

(Aspect A5)

(173) The manipulation part is configured to be externally manipulated above the housing.

(174) (Aspect A6)

(175) The manipulation part is configured to be externally manipulated on a lateral side of the housing.

(176) (Aspect A7)

(177) The upper plate part has a fan-like shape or an arcuate shape, and the outer peripheral part has an arcuately curved plate-like shape.

(Aspect A8)

(178) The housing has a body engagement groove in which an inner edge part of the upper plate part of the cover body can be fitted.

(179) Further, the present disclosure also provides the following Aspects B1 to B25, in order to achieve another non-limiting object to provide improvement relating to a structure for restricting movement of a cover of a tool accessory. Any one of Aspects B1 to B25 can be employed alone or two or more of them can be employed in combination. Alternatively, at least one of Aspects B1 to B25 can be employed in combination with at least one of the features of the grinders 1A, 1B, 1C, 1D, 1E, 1F, 1G of the above-described embodiments, the above-described modifications, Aspects A1 to A8 and the claimed features.

(180) (Aspect B1)

(181) A power tool, comprising: a spindle that (i) extends along a drive axis that defines an up-down direction of the power tool, and (ii) has a lower end portion that is configured such that a tool accessory is removably mounted thereto; a housing that houses the spindle with the lower end portion exposed to an outside of the housing; a cover body that is configured to be removably mounted to the housing to partially cover the tool accessory mounted to the lower end portion of

the spindle; and a movement restricting part that is provided on or in the cover body, wherein: the movement restricting part is configured to restrict (i) movement of the cover body in a circumferential direction around the drive axis relative to the housing and (ii) movement of the cover body in another direction that is different from the circumferential direction relative to the housing.

(182) In the power tool according to this aspect, the movement restricting part provided on/in the cover body can restrict the movement of the cover body relative to the housing (i) in the circumferential direction (i.e., rotation of the cover body) and (ii) in the other direction that is different from the circumferential direction. Thus, the movement restricting part can perform two functions of (i) properly positioning the cover body in the circumferential direction and (ii) suppressing looseness or play in the other direction. Therefore, compared with a case in which separate structures respectively perform these two functions, a space required in the power tool can be reduced and the power tool can be simplified in structure.

(183) (Aspect B2)

(184) The power tool as defined in Aspect B1, wherein: the cover body includes (i) an upper plate part that is configured to be arranged above the tool accessory mounted to the lower end portion of the spindle and (ii) an outer peripheral part that extends along an outer edge of the upper plate part and protrudes downward from the outer edge, and the movement restricting part is configured as a protruding part that protrudes radially outward relative to the drive axis from the upper plate part or the outer peripheral part.

(185) According to this aspect, the movement restricting part does not need to protrude downward from the cover body, and thus size increase of the power tool in the up-down direction can be avoided.

(186) (Aspect B3)

(187) The power tool as defined in Aspect B2, wherein: the housing has (i) a groove and (ii) a movable member that is movable between (a) an engagement position at which the movable member engages with the protruding part and (b) a disengagement position at which the movable member is not engageable with the protruding part, and the protruding part is configured to: restrict movement of the cover body in the other direction relative to the housing by being engaged with the groove, and restrict movement of the cover body in the circumferential direction relative to the housing by being engaged with the movable member when the movable member is at the engagement position.

(188) According to this aspect, restriction of the looseness in the other direction and positioning of the cover body in the circumferential direction can be effectively achieved by cooperation between the protruding part on the cover body, and the groove and the movable member that are provided on or in the housing.

(189) (Aspect B4)

(190) The power tool as defined in Aspect B2 or B3, further comprising: a movable member that is provided on or in the housing to be movable between an engagement position and a disengagement position, wherein: the movable member includes: a manipulation part that is configured to be externally manipulated above the housing by a user, an engagement part that is configured to engage with the protruding part of the cover body when the movable member is at the engagement position, and not to be engageable with the protruding part of the cover body when the movable member is at the disengagement position, and a connection part that connects the manipulation part and the engagement part.

(191) According to this aspect, the movable member is provided that is easy for a user to operate.

(192) (Aspect B5)

(193) The power tool as defined in Aspect B4, wherein the movable member is a lever that is linearly movable substantially in parallel to the drive axis.

(194) According to this aspect, the cover body can be positioned in the circumferential direction by

cooperation between the protruding part protruding from the cover body and the movable member (lever) having a simple structure.

(195) (Aspect B6)

(196) The power tool as defined in Aspect B5, wherein the connection part is at least partially within the housing.

(197) According to this aspect, a possibility that the movable member is affected by an unintended external force can be reduced.

(198) (Aspect B7)

(199) The power tool as defined in Aspect B5 or B6, wherein the engagement part is configured to engage with the protruding part of the cover body from below when the movable member is at the engagement position.

(200) According to this aspect, a user can move the lever from the engagement position to the disengagement position by pushing the manipulation part downward, so that the manipulation is easier compared with a structure that requires pulling the manipulation part upward.

(201) (Aspect B8)

(202) The power tool as defined in Aspect B2 or B3, further comprising: a lever that (i) includes a manipulation part that is configured to be externally manipulated by a user, and (ii) is provided on or in the housing to be pivotable between an engagement position at which the lever engages with the protruding part and a disengagement position at which the lever is not engageable with the protruding part.

(203) According to this aspect, restriction of the looseness in the other direction and positioning of the cover body in the circumferential direction can be effectively achieved by cooperation between the protruding part of the cover body and the easy-to-operate rotary (pivotable) lever.

(204) (Aspect B9)

(205) The power tool as defined in Aspect B8, wherein: the housing extends in a front-rear direction that is orthogonal to the drive axis, and the lever is configured such that the manipulation part extends in the front-rear direction when the lever is at the engagement position.

(206) According to this aspect, the manipulation part extends along the housing when the lever is at the engagement position, so that the lever does not easily interfere with user's operation.

(207) (Aspect B10)

(208) The power tool as defined in Aspect B8 or B9, wherein the lever is configured to be pivoted from the engagement position to the disengagement position when the manipulation part is pushed toward the housing.

(209) This aspect provides the lever that is easy to operate.

(210) (Aspect B11)

(211) The power tool as defined in Aspect B9 or B10, further comprising: a motor that is housed within the housing and configured to drive the spindle; a switch that is provided to start the motor and housed within the housing; and a switch manipulation member that is operably coupled to the switch and held by the housing so as to be externally manipulated by a user, wherein: the lever is configured such that a rear end of the manipulation part is located forward of the switch manipulation member when the lever is at the engagement position.

(212) According to this aspect, the lever does not easily interfere with user's operation of the switch manipulation member.

(213) (Aspect B12)

(214) The power tool as defined in any one of Aspect B2 to B11, wherein an upper surface of the protruding part is substantially flush with an upper surface of the upper plate part.

(215) According to this aspect, the upper surface of the protruding part and the upper surface of the upper plate part can be integrally utilized to restrict upward movement of the cover body (to suppress looseness in the up-down direction).

(216) (Aspect B13)

(217) The power tool as defined in any one of Aspect B2 to B12, further comprising: a movable member that is provided on or in the housing to be movable between an engagement position and a disengagement position, wherein: the protruding part is a rail having at least one recess or hole, and the movable member is configured to engage with the recess or the hole in the engagement position and not to be engageable with the recess or the hole in the disengagement position.

(218) According to this aspect, rotation of the cover body relative to the housing can be reliably restricted.

(219) (Aspect B14)

(220) The housing has an annular groove formed around the drive axis, the upper plate part of the cover body has a fan-like shape having an arcuate inner edge and an arcuate outer edge, and the upper plate part has an inner edge part that extends along the inner edge and is configured to be fitted in the annular groove.

(221) The inner edge **511**, the outer edge **515** and the inner edge part **512** of the upper plate part **51** are examples of an “inner edge”, an “outer edge” and an “inner edge part” of this aspect, respectively. The body engagement groove **311** is an example of an “annular groove” of this aspect.

(222) (Aspect B15)

(223) In Aspect B14, the protruding part is an arcuate rail, and the rail has an outer edge part that extends along an outer edge and is configured to be fitted in the groove of the housing.

(224) The outer edge part **556** is an example of an “outer edge part” of this aspect.

(225) According to this aspect, the upper plate part and the movement restricting part (protruding part) are fitted in the annular groove and the groove, respectively, so that movement of the cover body in the radial direction relative to the housing can be effectively suppressed.

(226) (Aspect B16)

(227) The power tool further includes a biasing member that is configured to bias the movable member toward the engagement position.

(228) According to this aspect, engagement between the movable member and the protruding part can be maintained by the biasing force of the biasing member.

(229) The biasing members **357A**, **357E** are examples of a “biasing member” of this aspect.

(230) (Aspect B17)

(231) The protruding part has at least one recess or hole, and the engagement part of the movable member (i) has a projection protruding upward and configured to engage with the recess or the hole, and (ii) is configured to be moved downward from the engagement position to the disengagement position relative to the protruding part of the cover body.

(Aspect B18)

(232) The movable member is configured to restrict movement of the cover body in the circumferential direction and/or movement of the cover body in the other direction relative to the housing by pressing the protruding part against the housing when the movable member is in the engagement position.

(233) (Aspect B19)

(234) In Aspect B18, the protruding part has at least one recess or projection, and the housing has a projection or recess that is configured to engage with the recess or the projection of the protruding part when the protruding part is pressed against the housing by the movable member.

(235) The engagement recess **56E** is an example of a “recess of the protruding part” of this aspect. The engagement projection **32E** is an example of a “projection of the housing” of this aspect.

(236) (Aspect B20)

(237) The power tool further includes a fan that is configured to be rotated integrally with an output shaft of the motor, and a rear end of the manipulation part is located forward of the fan when the lever is in the engagement position.

(Aspect B21)

(238) The housing includes a handle part that is configured to be held by a user, and a rear end of

the manipulation part is located forward of the handle part when the lever is in the engagement position.

(Aspect B22)

(239) The cover body is pivotable around the drive axis relative to the housing when the movable member is in the disengagement position, the protruding part has a plurality of recesses or holes, and the movable member is configured to be selectively engaged with one of the recesses or the holes when the movable member is in the engagement position.

(Aspect B23)

(240) The upper plate part has a fan-like shape or an arcuate shape, and the outer peripheral part is an arcuately curved plate-like part.

(Aspect B24)

(241) The protruding part is between an upper end and a lower end of the cover body in the up-down direction.

(242) The position “between the upper end and the lower end” in this aspect includes the same location as the upper end and the same location as the lower end.

(243) (Aspect B25)

(244) A lower end of the movable member is located in the same location as or above a lower end of the cover body in the up-down direction when the movable member is in the engagement position and in the disengagement position.

(245) Correspondences between the features of the above-described embodiments and the features of the above-described aspects B1 to B25 are as follows. However, the features of the above-described embodiments are merely exemplary and do not limit the features of the present disclosure or invention.

(246) The grinders **1A**, **1B**, **1E**, **1F** are examples of a “power tool”. The spindle **25** is an example of a “spindle”. The housings **10A**, **10B**, **10E**, **10F** are examples of a “housing”. The cover body **50** is an example of a “cover body”. The rails **55A**, **55E**, **55F** are examples of a “movement restricting part” and also a “protruding part”.

(247) The upper plate part **51** is an example of an “upper plate part”. The outer peripheral part **53** is an example of an “outer peripheral part”. The rail engagement grooves **32A**, **32F** are examples of a “groove”. The levers **35A**, **35B**, **35E**, **35F** are examples of a “movable member”. The manipulation part **351A** is an example of a “manipulation part”. The engagement parts **353A**, **353B** are examples of an “engagement part”. The connection part **355A** is an example of a “connection part”. The levers **35A**, **35B** are examples of a “lever” that is linearly movable. The levers **35E**, **35F** are examples of the a “lever” that is pivotable. The manipulation part **351E** is an example of the “manipulation part” of the lever that is pivotable. The motor **21** is an example of a “motor”. The switch **27** is an example of a “switch”. The switch knob **271** is an example of a “switch manipulation member”. The rails **55A**, **55E**, **55F** are examples of a “rail”. The engagement recess **56E** is an example of a “recess”. The engagement holes **56A**, **56F** are examples of a “hole”.

(248) The above-described embodiments are merely exemplary, and the power tool according to Aspects B1 to B25 of the present disclosure is not limited to the grinders **1A**, **1B**, **1E**, **1F** of the above-described embodiments. For example, the following modifications may be made. At least one of these modifications can be employed in combination with at least one of the features of the grinders **1A**, **1B**, **1E**, **1F** of the above-described embodiments and the claimed invention.

(249) For example, the present disclosure may be applied to other kinds of power tools (e.g., as a cutter, a circular saw and a multi-tool) configured such that a cover that is configured to partially cover a tool accessory mounted to a spindle is removably mounted thereto. The structure of the housing, elements (mechanisms) disposed within the housing, and/or their arrangement can be appropriately changed according to the kind of the power tool.

(250) The shape of the cover body **50** of the above-described embodiments (e.g., the length of the upper plate part **51** in the circumferential direction or the radial direction) can be appropriately

changed. The shapes (e.g., the length in the circumferential direction, the width in the radial direction, and/or the thickness in the up-down direction) of the rails **55A**, **55E**, **55F** and/or the positions of the rails **55A**, **55E**, **55F** relative to the cover body **50** can be appropriately changed. The sectional shapes, sizes and/or arrangement of the engagement holes **56A**, **56F** and the engagement recesses **56E** can be appropriately changed. Further, the cover body of the present disclosure need not have an arcuate rail. For example, a movable member (e.g., a lever) that is movable between the engagement position and the disengagement position may be provided on or in the housing, and at least one projection, recess or hole may be provided in the outer peripheral part of the cover body so as to engage with the movable member when the movable member is at the engagement position.

(251) The configurations of the rail engagement parts **30A**, **30B**, **30E**, **30F** can be appropriately changed. For example, the sectional shapes and/or arrangement of the rail engagement grooves **32A**, **32F** can be appropriately changed according to the change of the rails **55A**, **55E**, **55F**. The configurations (e.g., the shapes, arrangement, supporting manners) of the levers **35A**, **35B**, **35E**, **35F** can be appropriately changed in relation to the configuration of the cover body **50** (such as the recess or the hole of the rails **55A**, **55E**, **55F**).

DESCRIPTION OF THE REFERENCE NUMERALS

(252) **1A**, **1B**, **1C**, **1D**, **1E**, **1F**, **1G**: grinder, **10A**, **10B**, **10C**, **10D**, **10E**, **10F**, **10G**: housing, **101A**, **101C**: holding groove, **102A**, **102C**: cover part, **102A**, **103C**: spring receiving recess. **104**: lower surface, **105**: cylindrical part, **107**: support part, **11**: driving-mechanism housing part, **13**: motor housing part, **15**: handle part, **17**: controller housing part, **18**: battery mounting part, **21**: motor, **211**: stator, **215**: output shaft, **22**: fan, **23**: intermediate shaft, **230**: drive gear, **25**: spindle, **250**: driven gear, **253**: tool mounting part, **254**: lock nut, **27**: switch, **271**: switch knob, **272**: connecting member, **29**: controller, **30A**, **30B**, **30C**, **30D**, **30E**, **30F**: rail engagement part, **30G**: outer peripheral engagement part, **31A**: body engagement part, **311**: body engagement groove, **32A**, **32F**: rail engagement groove, **32E**: engagement projection, **33**: restriction wall, **35A**, **35B**, **35C**, **35D**, **35E**, **35F**: lever, **351A**, **351C**, **351D**, **351E**: manipulation part, **353A**, **353B**, **353C**, **353D**, **353E**, **353F**, **353G**: engagement part, **354A**, **354B**, **354C**, **354D**, **354F**: engagement projection, **354G**: hook, **355A**, **355C**, **355D**, **355E**: connection part, **355G**: arm, **356A**, **356D**: spring receiving projection, **356E**: arm part, **357A**, **357C**, **357D**, **357E**: biasing member, **358D**: screw, **358E**: support pin, **35G**: latch, **5A**, **5C**, **5D**, **5E**, **5F**, **5G**, **5H**, **5J**: cover, **50**: cover body, **51**: upper plate part, **511**: inner edge, **512**: inner edge part, **513**: upper surface, **515**: outer edge, **53**: outer peripheral part, **535**: projection, **54**: lip part, **541**: projection, **55A**, **55C**, **55D**, **55E**, **55F**: rail, **551**: upper surface, **552**: lower surface, **555**: outer edge, **556**: outer edge part, **56A**, **56F**: engagement hole, **56D**, **56E**: engagement recess, **91**: tool accessory, **93**: battery

Claims

1. A power tool, comprising: a spindle that (i) extends along a drive axis that defines an up-down direction of the power tool and (ii) has a lower end portion that is configured such that a tool accessory is removably mounted thereto; a housing that houses the spindle with the lower end portion exposed to an outside of the housing; a cover body that is (i) configured to be removably mounted to the housing to partially cover the tool accessory mounted to the lower end portion of the spindle, and (ii) includes (a) an upper plate part that is configured to be arranged above the tool accessory and (b) an outer peripheral part that extends along an outer edge of the upper plate part and protrudes downward from the outer edge; a first engagement part that is provided on or in the cover body; and a second engagement part that (i) is provided on or in the housing and (ii) is movable between (a) an engagement position at which the second engagement part engages with the first engagement part and (b) a disengagement position at which the second engagement part is not engageable with the first engagement part, wherein: the first engagement part and the second

engagement part are configured to restrict rotation of the cover body around the drive axis relative to the housing by engaging with each other, the first engagement part is (i) at a same location as the outer peripheral part or radially outward of the outer peripheral part in a radial direction that is orthogonal to the drive axis, and (ii) between an upper end and a lower end of the cover body in the up-down direction, and the first engagement part is configured as a protruding part that protrudes radially outward from the upper plate part or the outer peripheral part of the cover body.

2. The power tool as defined in claim 1, wherein the second engagement part is a lever that is linearly movable substantially in parallel to the drive axis.

3. The power tool as defined in claim 2, wherein: the protruding part has at least one recess or hole that extends in the up-down direction, and the second engagement part has a projection that is configured to engage with the at least one recess or the hole when the second engagement part is at the engagement position.

4. The power tool as defined in claim 3, wherein the lever includes (i) a manipulation part that is configured to be externally manipulated by a user above the housing, (ii) the projection, and (iii) a connection part that connects the manipulation part and the projection.

5. The power tool as defined in claim 4, wherein the connection part is at least partially within the housing.

6. The power tool as defined in claim 4, wherein the projection is configured to be engaged with the protruding part of the cover body from below when the lever is at the engagement position.

7. The power tool as defined in claim 1, wherein the second engagement part is a rotary lever that is configured to be externally manipulated by a user.

8. The power tool as defined in claim 7, wherein the second engagement part is movable within a plane that is substantially orthogonal to the drive axis.

9. The power tool as defined in claim 8, wherein: the protruding part has at least one recess that is recessed radially inward from an outer edge of the protruding part; and the second engagement part has a projection that is configured to be engaged with the recess when the second engagement part is at the engagement position.

10. The power tool as defined in claim 1, wherein the protruding part is configured to restrict movement of the cover body relative to the housing in a first direction that is different from a circumferential direction around the drive axis by engaging with one or both of the second engagement part and the housing.

11. The power tool as defined in claim 10, wherein: the housing has a groove, and the protruding part is configured to restrict movement of the cover body relative to the housing in the first direction by being fitted in the groove.

12. The power tool as defined in claim 10, wherein the second engagement part is configured to restrict (i) rotation of the cover body relative to the housing and (ii) movement of the cover body relative to the housing in the first direction by pressing the protruding part against the housing when the second engagement part is at the engagement position.

13. The power tool as defined in claim 1, wherein the protruding part is an arcuate rail.

14. The power tool as defined in claim 1, further comprising a biasing member that is configured to bias the second engagement part toward the engagement position.

15. The power tool as defined in claim 1, wherein the first engagement part is formed separately from the cover body and fixed to the cover body.

16. The power tool as defined in claim 1, wherein a lower end of the second engagement part is at a same location as or above a lower end of the cover body in the up-down direction when the second engagement part is at the engagement position and when the second engagement part is at the disengagement position.

17. The power tool as defined in claim 16, wherein an entirety of the second engagement part is between an upper end and the lower end of the cover body in the up-down direction when the second engagement part is at the engagement position and when the second engagement part is at

the disengagement position.

18. A power tool, comprising: a spindle that (i) extends along a drive axis that defines an up-down direction of the power tool, and (ii) has a lower end portion that is configured such that a tool accessory is removably mounted thereto; a housing that houses the spindle with the lower end portion exposed to an outside of the housing; a cover body that (i) is configured to be removably mounted to the housing to partially cover the tool accessory mounted to the lower end portion of the spindle, and (ii) includes (a) an upper plate part that is configured to be arranged above the tool accessory and (b) an outer peripheral part that extends along an outer edge of the upper plate part and protrudes downward from the outer edge; a rail that (i) has an arcuate shape, (ii) protrudes radially outward from the upper plate part or the outer peripheral part of the cover body, and (iii) is entirely located radially outward from the upper plate part or the outer peripheral part of the cover body; and a lever that (i) is configured to be externally manipulated by a user and (ii) is supported by the housing to be movable between (i) an engagement position at which the lever engages with the rail and (b) a disengagement position at which the lever is not engageable with the rail, wherein the rail and the lever are configured to restrict rotation of the cover body around the drive axis relative to the housing by engaging with each other.

19. The power tool as defined in claim 18, wherein the rail is configured to restrict movement of the cover body relative to the housing in a first direction that is different from a circumferential direction around the drive axis by being engaged with one or both of the lever and the housing.
